



# AYN AL-SIJILL

## Revenue can fail while the dashboard still looks healthy.

AYN AL-SIJILL reveals failures hidden between payment, order, and inventory systems. It gives the business one clear view of what happened, why it happened, and where action is needed.

### AYN AL-SIJILL

02 — CORRELATE

## One trace. One cause.

The team gets the context to act, not another dashboard to interpret.

trace.id / trc-8F21-A91D

14:22:07.114 PAYMENT\_SUCCESS

14:22:07.281 ORDER\_CREATE\_FAILED

14:22:07.283 DATABASE\_TIMEOUT

ROOT CAUSE / DATABASE TIMEOUT

AYN Incident Bot

ALERT DELIVERED NOW

HIGH

### Ghost Order detected

IMPACT Payment approved, order missing  
CAUSE Database connection timeout  
SERVICE order-service

OPEN MATCHING TRACE

#### THE BUSINESS PROBLEM

### The Ghost Order

A payment is authorized, but no order is created. This can lead to support cases, reversed authorizations, and lost customer trust. AYN AL-SIJILL connects the synthetic journey so teams can investigate the failure with context.

#### WHAT THE SOLUTION BRINGS TO THE BUSINESS

01

#### See hidden failures

Find issues that ordinary success metrics can miss.

02

#### Understand the cause

Connect the customer journey across every service.

03

#### Respond with context

Give teams the evidence they need to act quickly.

04

#### Learn across patterns

Use reporting data to reduce repeated incidents.

#### THE WIDER VIEW

### Checkout

SUCCESS AND FAILURE JOURNEYS

### Alerts

IMMEDIATE INCIDENT CONTEXT

### Reports

QUERYABLE REPORTING FOUNDATION

The demonstration uses synthetic data. It shows how the same approach can protect real customer journeys when connected to production systems with the right governance and security controls.

# Azure and the services we used.

AYN AL-SIJILL combines managed Azure services with a containerized observability stack. Each component has a focused role, from generating checkout journeys and protecting credentials to storing evidence, investigating incidents, and producing reports.

## MICROSOFT AZURE

### ● Azure Functions

Runs the synthetic checkout API, scheduled journeys, incident logic, and reporting endpoints.

### ● Azure Virtual Machine

Hosts the Dockerized Caddy and Elastic services used for ingestion and investigation.

### ● Azure SQL Database

Stores a structured reporting copy of operational events for later analysis.

### ● Azure Key Vault

Protects runtime secrets and the optional Telegram bot credentials.

### ● Azure Blob Storage

Stores remote Terraform state and supports state locking during infrastructure changes.

### ● Azure Logic Apps

Starts the observability VM every day at 09:00 Riyadh time.

### ● VM Auto-shutdown

Deallocates the VM at 23:00 Riyadh time to control demo operating costs.

### ● Azure networking

Allows public HTTPS while keeping SSH, Elasticsearch, Kibana, and Logstash data ports private.

## OBSERVABILITY SERVICES ON THE VM

### Caddy

HTTPS and routing

### Logstash

Event processing

### Elasticsearch

Evidence storage

### Kibana

Investigation views

### Filebeat

Linux and Docker logs

## PROVISIONING, DELIVERY, AND RESPONSE

### GitHub Actions

Defines validation and optional OIDC deployment.

### Terraform

Provisions Azure resources and keeps the environment repeatable.

### Cloud-init + Docker Compose

Prepares the VM and runs the service stack.

### Bash validation

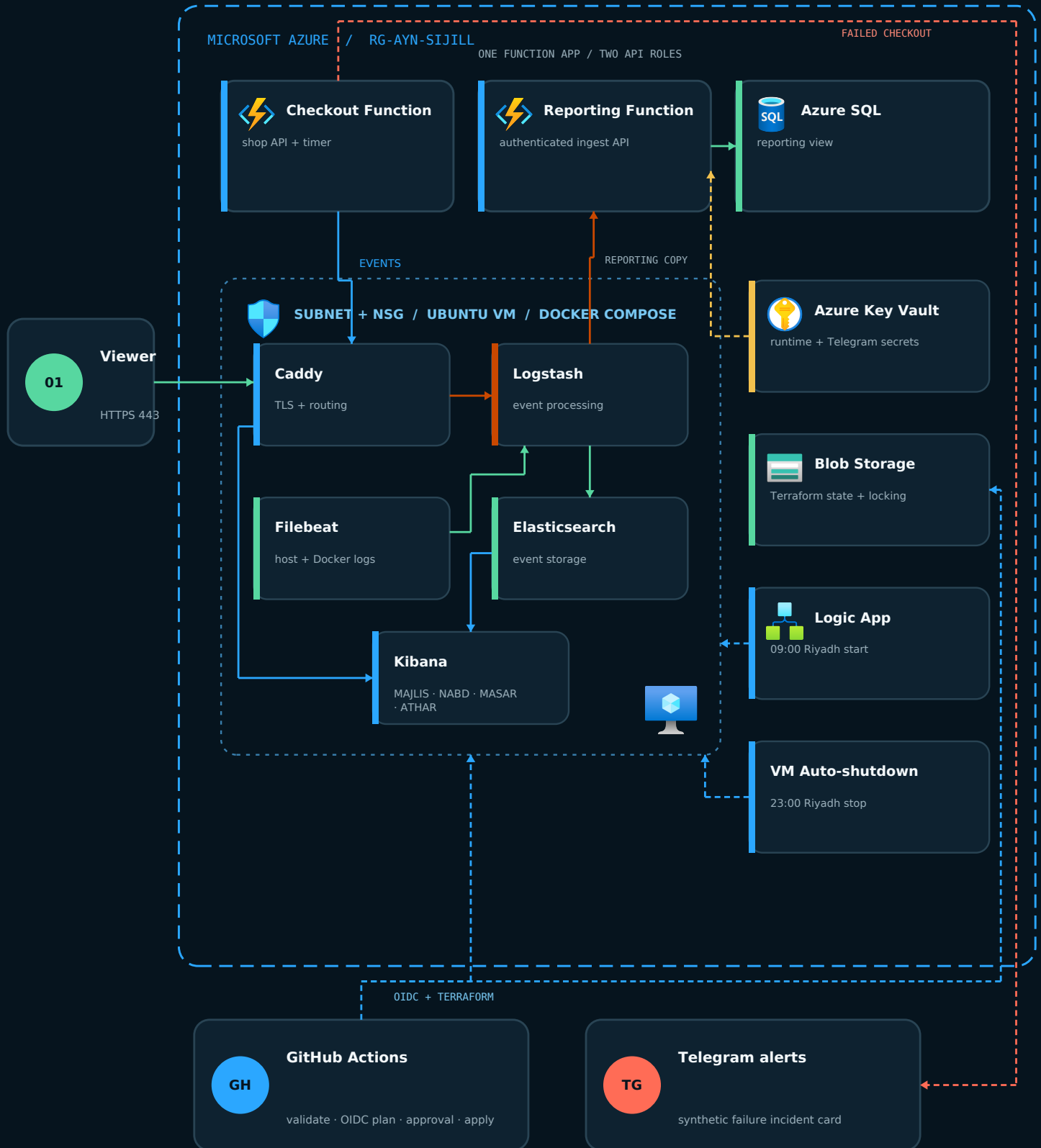
Configures, validates, and seeds historical journeys.

### Telegram Bot API

Sends alerts for failed checkout scenarios.

# One journey across the Azure platform.

The diagram shows the synthetic event path, SQL reporting branch, optional incident alerts, scheduled VM operations, and the GitHub Actions deployment workflow.



Solid lines: runtime data flow

Dashed lines: deployment, schedule, or secret access

Synthetic commerce data only

# Meet the team.

We met through the SDA Cloud Computing Bootcamp and built AYN AL-SIJILL by combining multidisciplinary thinking, business analysis, cybersecurity, and cloud delivery. Different perspectives helped us examine the same problem from the customer, business, security, and operational sides.

SDA - Cloud Computing Bootcamp

## Meet the team.

Explore our backgrounds, experience, and resumes.

S



THE ALL-ROUNDER

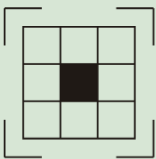
### Saud

Multidisciplinary

A jack-of-all-trades, connecting the dots across disciplines and bringing the bigger picture into focus.

 GitHub  LinkedIn Soon

N



THE SYSTEMS THINKER

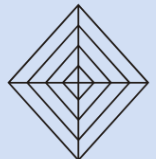
### Norah

IT & Business Analysis

Where technology meets business. An information systems background and a perspective that connects people, processes, and IT.

 GitHub Soon  LinkedIn Soon

R



THE SECURITY MIND

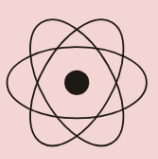
### Retaj

Cybersecurity

A cybersecurity perspective on a connected world. Bringing security into the conversation, from the first idea onward.

 GitHub Soon  LinkedIn Soon

L



THE CAREFUL OBSERVER

### Lama

Cybersecurity

Another lens on cybersecurity. A shared interest in building with care and understanding what makes a system secure.

 GitHub Soon  LinkedIn Soon

## FOUR COMPLEMENTARY PERSPECTIVES

**S** **Saud**  
Multidisciplinary

**N** **Norah**  
IT & Business Analysis

**R** **Retaj**  
Cybersecurity

**L** **Lama**  
Cybersecurity

## CONTINUE THE CONVERSATION

### Visit our team website.

Explore our backgrounds, experience, and resumes. The website is separate from the AYN AL-SIJILL solution and serves as the introduction to the people who created it.

[sda-team-collective.vercel.app](https://sda-team-collective.vercel.app)

Scan the code or select the address in the PDF.

